



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Branch: CSE (2025-29)

Semester: 3

Subject: Programming with Java

ASSIGNMENT No : 1

TITLE :Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

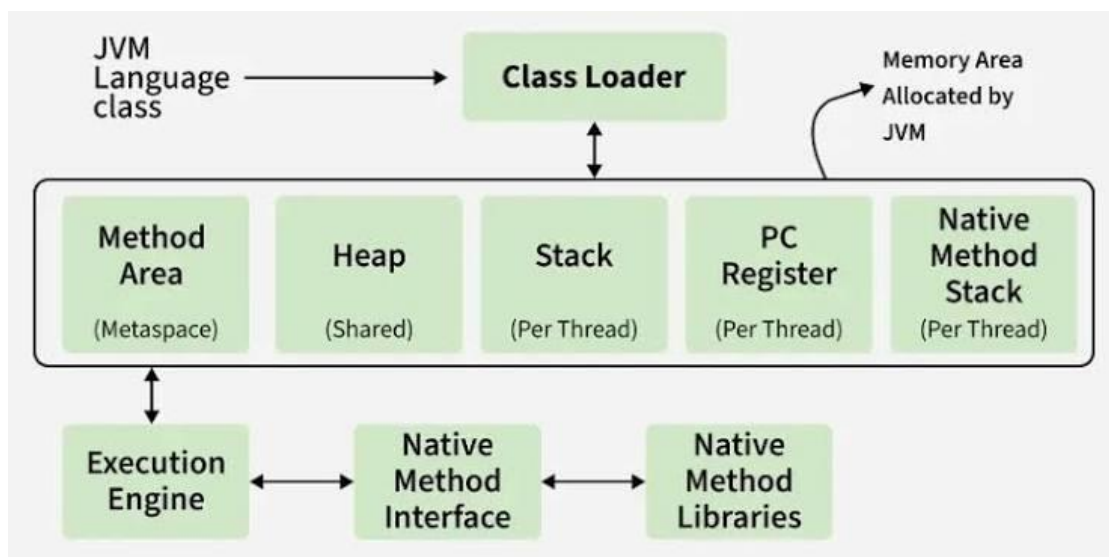
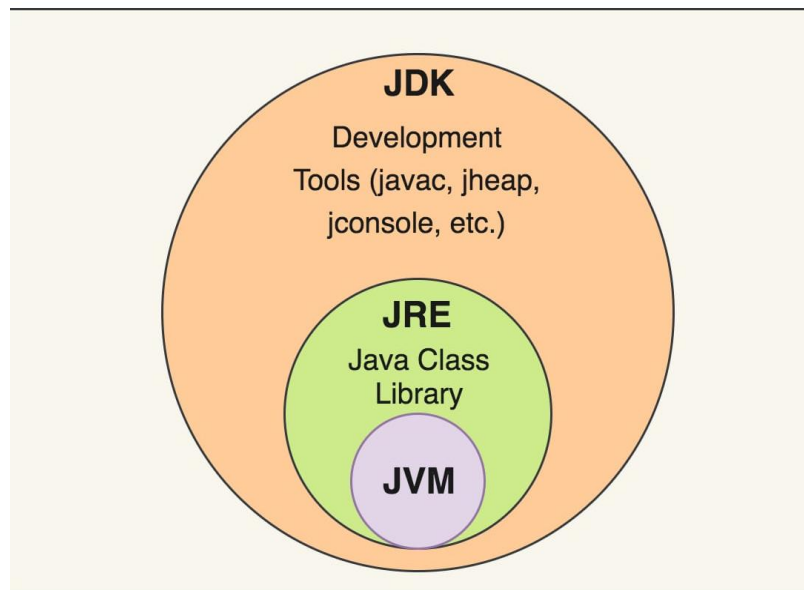
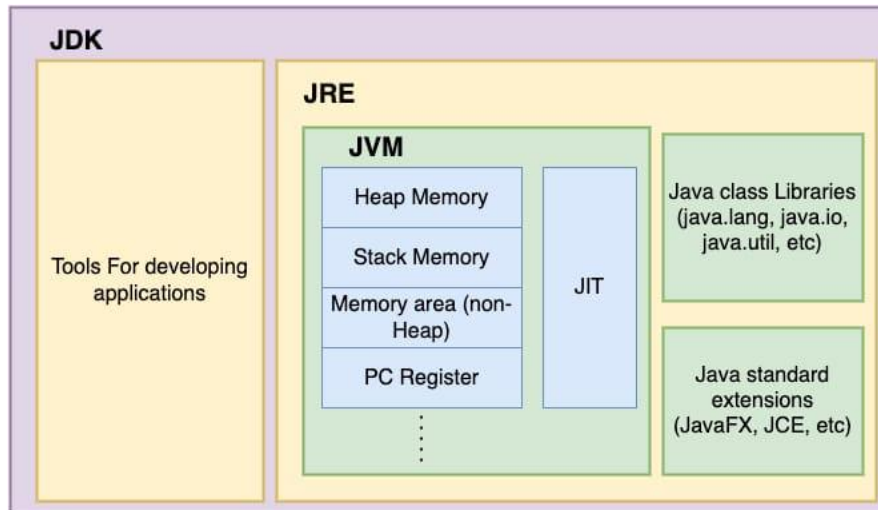
Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyze outputs, and apply the concepts to solve practical programming problems.

Theory

Java environment setup, JDK/JRE/JVM, variables, methods and constructors. forms an important component of Java programming. Students should understand the underlying concepts, architecture, advantages, use cases, and implementation methodology. The exercise should focus on understanding the design principles, coding practices, execution flow, and practical applications of these concepts in software development. Detailed study should include syntax, APIs, execution process, and real-world applications.

Output Screenshot



Conclusion:

The experiment helped us understand the basics of Java programming, including the Java environment setup, JDK, JRE, JVM, variables, methods, and constructors. By implementing these concepts in a simple program, I learned how Java applications are created, compiled, and executed. Likewise, it helped us understand writing organized and reusable code for developing java applications.

Questions:

1. Explain the main concept used in Java environment setup, variables, and methods.

Setting up the Java environment starts with installing the Java Development Kit (JDK), which includes all the tools required to create, compile, and run Java programs. After the JDK is installed, we can begin writing Java code. Variables are used to store different types of data, such as numbers, text, and characters. Methods are reusable blocks of code that perform specific tasks whenever they are called. In a Java program, variables hold the required data, while methods define the operations performed on that data. Together, they form the foundation of Java programming and help in building efficient and organized applications.

2. What are the advantages of using Java variables and methods?

Variables are used to store and manage data in a program, making the code more organized and easier to understand. Methods help divide a program into smaller, reusable sections, which reduces code repetition and improves readability. They also make debugging and maintaining the program simpler, as any changes only need to be made in one place. Since methods can be called multiple times with different inputs, they promote code reusability and save development time. Overall, variables and methods play an important role in creating well-structured, efficient, and easy-to-maintain Java programs.

3. How can this implementation be improved?

The program can be improved by taking user input, handling errors, and using better coding practices like encapsulation and multiple classes.

4. What are the real-world applications?

Java variables and methods are widely used in real-world applications. In banking systems, variables store account information, while methods handle tasks such as deposits and withdrawals. In e-commerce applications, variables manage product and customer data, and methods perform operations like adding items to the cart and processing payments. They are also used in mobile apps, enterprise software, and automation systems to organize data and implement reusable functionality efficiently.